From Commodore 64 to Coding Agents

What one person actually does with it

William Gerken

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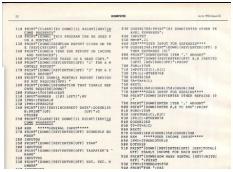
Research Excellence Series: Leveraging AI for Research
October 30, 2026

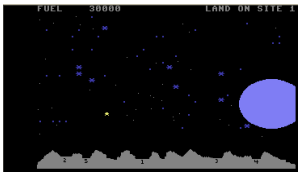
Christmas, with a Commodore 64





COMPUTE!, December 1983





Typed in, one line at a time. Lunar Lander.

A typo at five taught me the first rule: **the computer does what you tell it**, not what you mean.

Meaning versus language



Computer engineering, WVU

- C++ and Java were the cool languages. **Ada taught abstraction.**
- Loops, types, procedures: structures that are themselves abstractions of ones and zeros, which are transistors switching on and off. Underneath, logic gates. We built those too.
- C and C++ came easily after. I'd learned the level, not the syntax.



My view: a new interface to code, and the old rules apply

- Generative AI is one more layer: **human language as the interface to code**.
- The rules of abstraction still hold. Human language is the trickiest layer yet, because it is imprecise.
- A compiler does the same thing every time. An agent doesn't.

Dijkstra, 1978, on "natural language programming"

"A change of interface can easily increase at both sides of the fence the amount of work to be done."

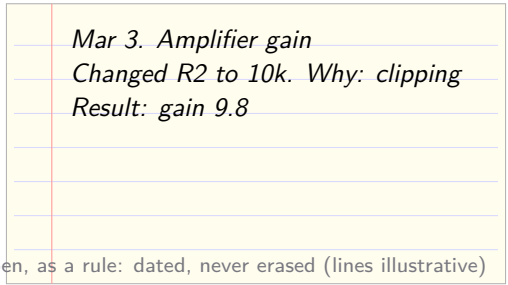
Make the instructions **precise enough to check**. My answer is old-fashioned: write it down.

Horses for courses



- **Claude Code**, the agent: **lives in my folders**, runs my code, keeps the notebook
- **Copilot in Outlook**: mail and calendar, on the university's license
- **A DGX Spark in my office**, running gpt-oss, Llama and Qwen locally: for text that stays in the building
- **ChatGPT**: speech and images

In engineering I kept a lab notebook. Now I keep folders.



in pen, as a rule: dated, never erased (lines illustrative)

A real folder of mine: the 2012 paper's tables

```
New_Tables_20_Oct_2010.xlsx
New_Tables_20_Oct_2010 (2).xlsx
New_Tables_27_Oct_2010.xlsx
New_Tables_01_Dec_2010.xlsx
New_Tables_05_Dec_2010.xlsx
New_Tables_28June2011.xlsx
```

You already have **more than you think**. You just can't find it.

The virtual notebook: four pages, all plain text

Page	What it holds	One real entry
Literature	60 papers indexed, read in full	Egan, Matvos and Seru, *JPE* 2019: 7% of advisers have a misconduct record. PDF in an old project folder.
Data	260 folders, with the terms	Keeneland auction catalogs, 2002–2025, public, parsed from PDF. Catch: about one page in eight does not parse.
Skills	29 plain-English procedures	Verify the paper: trace every table cell to the code that made it; flag any mismatch.
Decisions	471 diary entries	The brokerage data provider is never named. Every draft says **"a large discount brokerage."**

When I correct it, **the correction stays corrected.**

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- **Memory:** corrections that stay corrected. A schedule: every Monday, what's waiting on me.

Generative AI is probabilistic. **Everything here is about repeatability.**

Guardrails: rules it reads, hooks that run

14 rules: plain-English pages it reads every session

- Every number names its source, or says "illustrative"
- A Stata result is spot-checked in R or Python before it is reported
- On my prose: comments, not rewrites
- Restricted data: the code that touches it runs where the agent can't reach

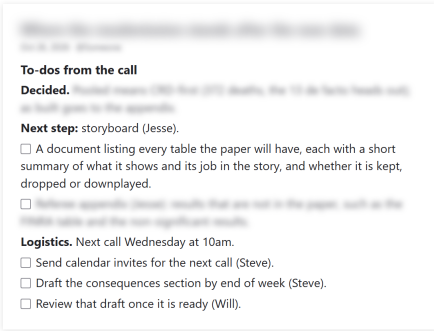
Hooks: small programs that run on every save or command, and **block** the risky ones

- No private folder enters the saved history
- No mass delete outside the project folder
- No uploads, no reading of credentials

Why the delete check exists

In October a cleanup script wiped my Temp folder. The lesson went in as a rule that afternoon, and as a check that evening.

Coauthors, who did not sign up for any of this



To-dos from the call

Decided. [illegible]

Next step: storyboard (Jesse).

- ☐ A document listing every table the paper will have, each with a short summary of what it shows and its job in the story, and whether it is kept, dropped or downplayed.
- ☐ [illegible]

Logistics. Next call Wednesday at 10am.

- ☐ Send calendar invites for the next call (Steve).
- ☐ Draft the consequences section by end of week (Steve).
- ☐ Review that draft once it is ready (Will).

- **A call memo kept as a Claude Doc**, a shared artifact: coauthors comment in the margin; the agent answers in the thread or changes the section.
- **A coauthor who lives in Word.** A script reads the tracked changes and comments; I answer; it writes my replies back as tracked edits.

They never see the agent. They see a doc, and a reply.

The case: a 2012 paper, and a 2017 extension



Predicting fraud by investment managers ☆

Stephen G. Dimmock [a], William C. Gerken [b] Show more

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Abstract

We test the predictability of investment fraud using a panel of mandatory disclosures filed with the SEC. We find that disclosures related to past regulatory and legal violations, conflicts of interest, and monitoring have significant power to predict fraud. Avoiding the 5% of firms with the highest ex ante predicted fraud risk would allow an investor to avoid 29% of fraud cases and over 40% of the total dollar losses from fraud. We find no evidence that investors receive compensation for fraud risk through superior performance or lower fees. We examine the barriers to implementing fraud prediction models and suggest changes to the SEC's data access policies that could benefit investors.

Dimmock and Gerken, *Journal of Financial Economics*, 2012.

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2017, the handbook chapter with Joe Farizo: an RA read **1,745 SEC releases** by hand, from written directions, to extend the fraud labels.

One job, three generations

	2012, the paper	**2017, the handbook**	**2026, the agent**
Who read the releases	we did	one RA, by hand	the agent, ten at a time
Pace	slow	about 11 an hour	**all 1,745 in 35 minutes**
Time on the reading	years	lots of RA hours	about 10 agent-hours
Code	61 scripts, about 22,000 lines, 2008–2011	24 scripts, about 8,000 lines	47 scripts, about 3,000 lines, two weeks

2012 is the whole paper; 2026 is the extension only. Same written directions in 2017 and 2026.

It started from the notebook

```
## Directions (verbatim from the 2017 RA instructions, J.
Farizo, Sep 4, 2017)
After opening the PDF file, it is best to first search for
the term "registered". This will immediately pick up terms
"never registered" or "unregistered." If these terms are
used to describe the defendant firm (the firm where the
fraud occurred), then record the file name in the
spreadsheet and move on to the next one.
```

It read my rules, the data list, and the 2017 directions before touching a release.

It checked its own work against last time

one coded row, of 1,745

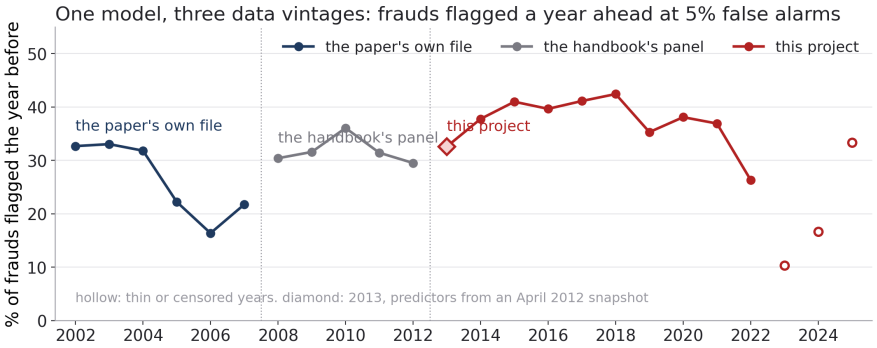
release: [redacted]
firm: [redacted]
registered: Y (Both)
class: Overstate assets
years: 2011-2015
amount: 900000
confidence: High
evidence: "Since 2008, [redacted] [redacted] has been registered as an investment adviser with the Commission."

- Reran my 2012 paper's main table: **every coefficient matches**.
- Agreed with the 2017 RA on **85 of 100**. The rest came to me.
- Coded 2017–2026 (3,390 documents) and ran the 2012 model on them, no refit: **still flags 27% of frauds a year ahead**, 2.7% false alarms.

What it caught along the way

- The 2017 instructions were a shortcut on the 2012 method. It read both and found **639 documents** the shortcut had dropped. I had them coded that afternoon.
- A wrong column letter in my own predictor build, found because the series didn't match the 2017 panel.
- One year with no usable data file. It built the year from the raw filing archive and labeled it a late snapshot.
- The cases it couldn't match to a registration: it told me which way the bias runs.

Does the 2012 model still work? 2002 to 2025



Three data vintages, one model, twenty years. Where the fraud labels are complete, **about a third of frauds or more**, every year.

“Doesn’t it make up citations?”

1. It cites from **my index**, not from its memory
2. It checks references against Crossref, the registry behind DOIs
3. What it can’t verify, it flags, never fills in

June audit: six bibliographies, about 370 references

No invented references. 18 real ones had stale details, in our own bibliographies.

Where it helps, and where it doesn't

Helps

- Finding what I've already read
- Writing and checking code
- Cleaning and documenting data
- **Pushing back on me**, if I ask

Doesn't

- Deciding which question matters
- Judging whether a design is credible
- **Verifying its own work**: that's on me

The risk isn't that it's slow. It's **getting out over your skis**.

What changed for me

- **A stalled project came back:** the data work got cheap
- **A shelved question got asked:** every Keeneland horse since 2002
- **"Can you check whether. . . ?"** is an afternoon. So is **"this won't work"**

It changes **what I spend time on**, not whether I think.

Start today: three files

1. **Standing instructions:** who you are, how you work
2. **A list of your data, with the terms**
3. **An index of what you've read:** point it at old folders

Mine grew over six months. **These three take an afternoon** (my estimate).

You already have more than you think.

Write it down where the agent can read it.

Then check what it does.

Questions and discussion

Additional Slides

Chat, or agent

	A chat window	**An agent**
Where it runs	a browser tab	**your project folder**
What it keeps	this chat	what you wrote down, read every session
What it can do	answer	read your files, run your code, check the output

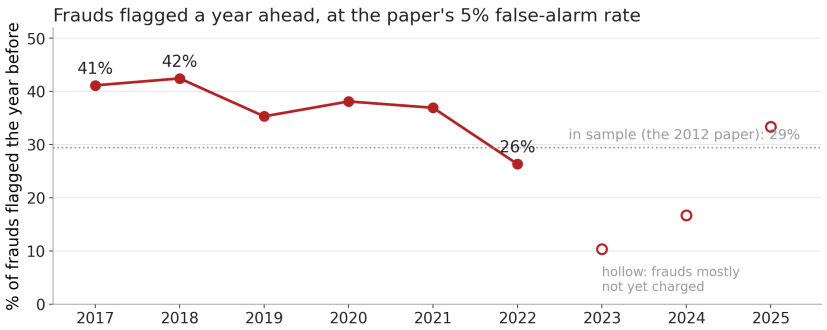
Where to learn more

If you have...	Start with
1 hour	Goldsmith-Pinkham, *Claude Code for Applied Economists*, Ep. 1
A weekend	Cunningham, *Scott's Mixtape* Substack
A semester	*The Mixtape with Scott* podcast
Reference	Korinek (2023), *JEL*; Velikov's wiki

Slides and handout: **willgerken.com/rexai.pdf**

Kristine Hankins' list: 13 resources, sorted by use

Year by year, 2017 to 2025



At the paper's own 5% false-alarm rate: **39% of 2017–2022 frauds** flagged a year ahead, against 29% in sample. Ranking quality (AUC) 0.80 in 2017 to 0.71 in 2022.

Citation audit, June 2026

Bibliographies checked	6
References, about	370
Real, with stale details (volume, issue, now published)	18
Worth a second look	6
Could not verify: flagged, not filled in	4
Invented	**0**

Targeted check: recent, forthcoming and inconsistently cited entries queried against Crossref; long-standing classics spot-checked.

A recipe: compile the paper

1. Find the paper's main file
2. Run LaTeX, then BibTeX, then LaTeX twice; never stop to ask
3. Read the log: report every error with its file and line
4. List any missing citations or references
5. Report: "compiled, N pages, M warnings," or the errors

Plain English, about twenty lines. I have 29 like it.

The three files

Standing instructions
Who you are and your field. How you work. What never to touch. How to check work.

A list of your data
Dataset, where it lives, vintage, shareable and why, the terms, the catch.

A literature index
"Read the papers in these folders. For each: finding, method, open questions. Only papers you actually read."